

PETROPAVLOVSK-KAMCHATSKY, Russian Federation

WMO: 325400

Lat: 53.083N

Lon: 158.583E

Elev: 84

StdP: 100.32

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.8	(c) -20.0	(d) -27.9	(e) 0.3	(f) -20.1	(g) -25.3	(h) 0.4	(i) -18.5	(j) 14.0	(k) 0.5	(l) 12.0	(m) -1.0	(n) 3.8	(o) 0	(p) 0.733

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.9	(c) 25.0	(d) 16.9	(e) 22.9	(f) 15.8	(g) 20.9	(h) 15.1	(i) 18.1	(j) 23.1	(k) 16.7	(l) 21.3	(m) 15.5	(n) 19.6	(o) 4.1	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.0	(b) 11.5	(c) 20.8	(d) 14.9	(e) 10.7	(f) 19.0	(g) 13.8	(h) 9.9	(i) 17.5	(j) 51.4	(k) 23.1	(l) 47.1	(m) 21.4	(n) 43.6	(o) 19.7	(p) 24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.8	(b) 9.2	(c) 7.8	(e) -24.8	(f) 28.5	(g) 2.6	(h) 2.2	(i) -26.6	(j) 30.1	(k) -28.1	(l) 31.4	(m) -29.6	(n) 32.6	(o) -31.5	(p) 34.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-9.5	-8.6	-5.1	0.0	5.5	10.8	14.5	14.7	10.7	4.3	-3.0	-8.4		
	DBStd	9.55	5.58	5.11	4.17	2.97	2.80	3.39	3.06	2.34	2.83	3.04	3.94	5.08		
	HDD10.0	3230	606	520	468	299	143	28	2	1	24	178	390	571		
	HDD18.3	5895	864	753	727	549	398	227	126	116	231	436	640	829		
	CDD10.0	388	0	0	0	0	4	53	140	146	44	1	0	0		
	CDD18.3	10	0	0	0	0	0	1	6	3	0	0	0	0		
	CDH23.3	137	0	0	0	0	1	31	71	32	2	0	0	0		
	CDH26.7	17	0	0	0	0	0	5	11	2	0	0	0	0		
(14) Wind	WSAvg	3.4	3.7	3.8	3.7	3.4	3.5	3.3	2.9	2.8	3.1	3.3	3.6	3.8		
(15) Precipitation	PrecAvg	1012	78	64	76	78	55	66	66	101	96	139	116	83		
	PrecMax	1330	167	245	179	173	113	175	163	211	161	376	367	314		
	PrecMin	719	13	11	16	20	18	15	6	30	20	38	22	17		
	PrecStd	177	42	49	37	44	24	41	36	47	36	77	75	64		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.2	4.1	10.8	19.2	27.0	28.1	26.2	22.9	14.8	7.1	4.2		
		MCWB	1.0	0.7	0.7	5.1	10.3	16.2	18.2	18.7	14.4	8.1	5.4	1.7		
	2%	DB	2.0	1.9	2.9	7.9	16.8	23.1	25.2	24.0	20.0	12.6	5.2	2.9		
		MCWB	-0.2	-0.2	0.1	3.2	9.4	14.1	17.4	16.8	13.2	7.4	3.4	0.8		
	5%	DB	0.8	0.4	1.9	6.0	14.0	20.8	23.0	22.0	18.0	10.9	4.1	1.2		
		MCWB	-1.0	-1.3	-0.5	2.1	7.9	13.7	16.5	16.0	12.0	6.4	2.3	-0.2		
	10%	DB	-0.6	-0.8	0.9	4.2	11.2	18.0	20.8	20.1	16.1	9.2	3.0	-0.1		
		MCWB	-2.0	-2.4	-1.0	1.0	6.4	12.1	15.8	15.0	11.1	5.8	1.1	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.0	1.4	5.7	11.6	17.4	19.7	20.4	16.0	10.0	6.1	2.6		
		MCDB	2.7	2.6	2.7	9.1	18.1	24.4	25.1	24.4	19.5	11.6	6.5	3.6		
	2%	WB	0.0	0.0	0.6	3.5	9.8	15.2	18.3	18.2	14.2	8.6	4.1	0.9		
		MCDB	1.1	1.3	1.8	6.9	15.9	21.5	23.8	22.4	18.4	10.7	4.8	2.0		
	5%	WB	-0.9	-1.0	-0.1	2.4	8.2	13.7	17.1	16.7	13.0	7.3	2.7	-0.1		
		MCDB	0.4	0.2	1.1	5.2	13.2	19.6	22.0	20.6	16.3	9.4	3.7	1.1		
	10%	WB	-2.2	-2.3	-0.8	1.5	6.7	12.4	15.7	15.5	12.0	6.4	1.1	-1.6		
		MCDB	-0.5	-0.7	0.5	3.8	10.8	17.6	20.1	19.1	15.1	8.6	2.5	-0.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	8.9	7.5	9.0	9.8	9.1	8.9	10.1	8.8	8.0	8.1		
		MCDBR	5.9	5.7	7.4	9.6	14.2	15.6	14.1	13.1	13.0	10.5	7.6	6.9		
		MCWBR	5.4	4.7	5.5	6.4	8.3	8.2	7.4	7.0	7.2	6.4	6.2	6.3		
		MCDBR	6.6	5.3	5.9	8.4	13.1	14.2	12.6	11.4	10.6	8.7	6.6	6.8		
	5% WB	MCWBR	6.2	4.7	4.8	5.9	8.0	8.1	7.4	7.0	6.6	6.2	6.2	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.341	0.403	0.462	0.515	0.492	0.495	0.454	0.380	0.367	0.326	0.300		
	taud		2.050	1.997	1.845	1.754	1.757	1.969	2.011	2.132	2.324	2.192	2.098	2.040		
	Ebn at Noon		624	721	747	755	739	766	755	765	788	696	594	547		
	Edh at Noon		76	111	160	199	210	171	162	136	99	91	72	64		
(44) All-Sky Solar Radiation	RadAvg		0.81	1.58	2.80	4.25	4.82	5.08	4.84	4.05	3.26	1.98	1.05	0.65		
	RadStd		0.10	0.23	0.29	0.48	0.47	0.61	0.45	0.40	0.29	0.20	0.10	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page